Group Project I: AI-Assisted Credit Card Fraud Detection

Goal

Build a model that flags fraudulent credit card transactions, **using an AI chatbot for every step**: loading the data, EDA, cleaning, modeling, the performance report, tuning, explanation, and the final README and slides. Work the way we did in class in `2026_11_05/ai_data_analytics.ipynb`: *less code, more chat*.

You are graded on what the AI produced **and on how well you checked it**. A chatbot writes plausible code quickly; your job is to give it clear instructions, read what it returns, catch its mistakes, and explain the results.

Data

`credit_card_fraud_10k.csv`: 10,000 card transactions, 10 columns, one row per transaction.

column	meaning
`transaction_id`	row identifier (not a feature)
`amount`	transaction amount
`transaction_hour`	hour of day, 0–23
`merchant_category`	Clothing, Electronics, Food, Grocery or Travel
`foreign_transaction`	1 if the transaction was abroad
`location_mismatch`	1 if the location does not match the cardholder's usual location
`device_trust_score`	trust score of the device used, 25–99
`velocity_last_24h`	number of transactions by the card in the last 24 hours
`cardholder_age`	cardholder's age

column	meaning
`is_fraud`	**target**: 1 = fraud

Only **151 of the 10,000 transactions (1.5%) are fraud**. This one fact shapes most of the project: read the "watch out" notes in each step.

Setup

1. Open the project folder in VS Code and start a **fresh chat session** (`Ctrl+Shift+P` → "Reload Window").
2. Create one new notebook for the group, `fraud_project.ipynb`.
3. For every step, put the prompt you gave the chatbot in a **markdown cell headed `Chat:`**, followed by the code cell(s) it produced, as in the class notebook. If you had to re-prompt or fix the output, keep the follow-up prompts too.

Steps

Each step lists what to ask for and what to check. Write your own prompts; the examples are starting points.

1. Load the data

Example prompt: "Load `credit_card_fraud_10k.csv` into a pandas DataFrame and report its shape, column types, missing values and duplicate rows."

Check: the row and column counts match the table above, and the fraud count is 151.

2. Exploratory data analysis

Example prompt: "Explore how fraud relates to each feature with seaborn: fraud counts, fraud rate by merchant category and by hour of day, and the distributions of amount, device trust score and velocity for fraud versus non-fraud."

Check:

- Fraud **rates**, not raw counts, are plotted. With 1.5% fraud, count plots hide the pattern.
- Your group can state in plain words which features separate fraud from legitimate transactions and which do not.

3. Clean and prepare

Example prompt: "Prepare the data for modeling: drop the identifier, encode the categorical column, and split into stratified training and test sets (80/20). Standardize numeric features, fitting the scaler on the training set only."

Watch out:

- `transaction_id` must not be a feature.
- The split must be **stratified**, or the test set may get very few frauds. Even stratified, the test set holds only about **30 frauds**, so a single test score is noisy. Also report **5-fold stratified cross-validation** on the training set.
- Any scaling, encoding, or resampling must be fitted **inside the training data only** (use a scikit-learn `Pipeline`). Ask the AI to show where each step is fitted.

4. Build models

Fit and compare at least four classifiers, for example:

1. logistic regression,
2. KNN or SVC,
3. random forest,
4. gradient boosting (scikit-learn, XGBoost or LightGBM).
5. or your best choice

Ask the AI how each model handles class imbalance (`class_weight="balanced"`, sample weights, or resampling) and use one approach consistently.

5. Report model performance

Example prompt: "Create a summary table of all models with accuracy, balanced accuracy, fraud precision, fraud recall, fraud F1, ROC AUC and PR AUC (average precision), on the test set and in 5-fold cross-validation."

Watch out:

- **Accuracy is misleading here.** A model that never predicts fraud scores 98.5%.
- **ROC AUC can look excellent while the model is weak at catching fraud.** Compare ROC AUC with PR AUC for each model, and explain the difference in your report.
- Show the **confusion matrix** of your best model and let AI plot **precision–recall curve**

6. Tune

1. **Threshold:** the default 0.5 cut-off is rarely right for rare events. Choose the threshold on a **validation split of the training data** (not the test set), maximizing F1 or meeting a business rule such as "catch at least 90% of fraud". Then report the test results once at that threshold.
2. **Hyperparameters:** tune your best model with Optuna (or grid search) using cross-validation on the training set.
3. Optionally, try an ensemble of your models, as in class, and report whether it helps.

Check: the test set was used only for the final evaluation. Ask the AI directly: "Did any step in this notebook use the test set for fitting or for choosing a setting?" Then verify its answer yourself.

7. Explain the model

Example prompt: "Use SHAP to explain the tuned model,e.g. global importance plot, beeswarm plot, or waterfall plot, and explanations for a correctly caught fraud and one missed fraud."

Interpret:

- Which features push predictions toward fraud, and in which direction?
- Do features **interact**? For example, is a foreign transaction riskier at night or with a low device-trust score? Compare how well logistic regression (no interactions unless you want to add them like x1*x2) does against the tree models, and explain the gap.
- If a model scores almost perfectly, do not celebrate: you may have data leakage issue (rerun without each feature in turn) and discuss whether this dataset is

realistic. Real fraud data is rarely this separable.

8. Write up

Ask the AI to draft, then edit them yourselves:

1. a **conclusion** cell: the best model, its test and cross-validation metrics, the threshold you chose and why, and the main fraud drivers from SHAP;
2. a **README.md**: purpose, setup, and how to run the notebook;
3. **presentation slides**: at most 5 slides for a 10-minute talk, focusing on the results.

Deliverables

1. `group_*_fraud_project.ipynb`: runs top to bottom, with the `Chat:` prompt cells, add your comments on AI generated code to show your understading.
2. create a `README.md`.
3. group presentation slides (`.pptx` or PDF), at most 5.
4. The **exported chat history** (`Ctrl+Shift+P` → "Chat: Export Chat…" as JSON, or right-click → Copy All into a txt or `.md` file).

Grading (10 points)

part	weight
Correct workflow: no leakage, stratified split, cross-validation, optimize threshold on validation, finetuning parameters	30%
Evaluate model performance: Model comparison and metrics appropriate for imbalanced data	30%
Explain model performance : SHAP, interactions, discussion of limitations	10%
Deliveries: README, slides, your comments in the notebook	10%
Group Presentation (in person)	20%